PhD Proposal

Causal Structure Learning in High Frequency Financial Markets

Nonparametric Local Independence Testing for Jump Diffusion Systems

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1 Project Scope

A recurring challenge in empirical finance is that the relationships we observe between variables are not necessarily causal. Two series can move together because one drives the other, because they share a common driver, or simply by coincidence. In most empirical settings, standard tools like regression or Granger causality tests can go some way toward addressing this, but they rely on strong functional form assumptions and are not well suited to continuous time systems where discrete events interact with continuous processes.

This is precisely the situation in high-frequency financial markets. Log-prices evolve as continuous diffusions, but they are constantly being shaped by discrete events: order arrivals, trade executions, policy announcements. The question of which of these events *genuinely drive* price dynamics as opposed to being correlated with them for other reasons is one that existing methods struggle to answer cleanly.

The formal framework for this kind of question is the *Local Independence Graph* (LIG) of Mogensen and Hansen (2020), a continuous-time causal graph that encodes which processes genuinely influence others after conditioning on the full available information set. The only currently available statistical test for local independence within this framework the cross fitted Local Covariance Test (X-LCT) of Christgau et al. (2023)works well, but it is restricted to settings where the *target* variable is a counting process. When the target is a continuous Itô diffusion, which is the case for log-prices, the test is not valid. This gap is the starting point for my project.

I want to extend the X-LCT to diffusion targets in mixed systems where counting processes and diffusions interact. If this can be done, it opens the door to a set of empirical questions in market microstructure, FX dynamics, and commodities that currently cannot be answered with the tools available. My research question is therefore:

Can the local independence testing framework be extended to Itô diffusion targets in mixed jump-diffusion systems, and what empirical questions does this make tractable?

I think the answer is yes, but the extension is not straightforward and involves resolving some non-trivial technical problems. I outline these below alongside a plausible path forward and the empirical questions I hope to pursue if the theory can be made to work.

2 Literature Review and the Gap I Want to Fill

2.1 Local independence graphs

Mogensen and Hansen (2020) develop the theory of Local Independence Graphs as a continuous-time analogue of the Bayesian network framework for causal discovery. Let

$Z = (Z^1, \ldots, Z^d)$ be a d-dimensional càdlàg process on a filtered probability space. Write $\mathcal{F}_t^A = \sigma(Z_s^i : i \in A,\, s \le t)$ for the filtration generated by a subset of coordinates, and $\mathcal{F}_t^{-\alpha}$ for the filtration generated by everything except Z^α.

Definition 2.1 (Mogensen and Hansen 2020)**.** Z^β is *conditionally locally independent* of Z^α given $Z^{-\alpha\beta}$, written $Z^\beta \perp\!\!\!\perp_{\mathrm{loc}} Z^\alpha \mid Z^{-\alpha\beta}$, if the $\mathcal{F}^{-\alpha}$-predictable compensator of Z^β equals its full $\mathcal{F}^{\mathrm{all}}$-compensator. In other words, knowing the history of Z^α adds nothing to predicting the evolution of Z^β once we already condition on everything else.

A directed edge $\alpha \to \beta$ is present in the LIG if and only if this local independence fails. Bidirected edges represent dependence that cannot be explained by the observed processes alone, pointing to a latent common driver. Mogensen and Hansen (2020) show that LIGs are closed under marginalisation over latent processes and admit a unique maximal Markov-equivalent representative, which is what makes causal discovery from observational data feasible in principle. Mogensen (2025) extends these identifiability results to a weaker notion of equivalence.

2.2 The existing test and why it does not cover my setting

The X-LCT of Christgau et al. (2023) is, to my knowledge, the only practically implementable test for conditional local independence. It is built around the *Local Covariance Measure* (LCM),

$$\gamma_t = \mathbb{E}\left[\int_0^t G_s \,\mathrm{d}M_s\right], \tag{1}$$

where $M_t = N_t - \int_0^t \lambda_s \,\mathrm{d}s$ is the compensated martingale of a counting process N, and $G_t = X_t - \Pi_t$ is a residual after projecting X onto the smaller filtration $\mathcal{F}^{-\alpha}$. Under the null of conditional local independence, $\gamma_t = 0$ for all t. A K-fold cross-fitted estimator of γ_t satisfies a Brownian bridge limit, which gives exact asymptotic p-values without needing to simulate critical values.

The test is elegant and works well for counting-process targets. The problem is that its validity rests on the identity $[M]_t = N_t$, which is specific to counting processes and fails entirely for Itô diffusions. In the canonical high frequency model,

$$\mathrm{d}X_t^i = \mu^i(t, X_t, N_t)\,\mathrm{d}t + \sigma^i(t, X_t, N_t)\,\mathrm{d}W_t^i + \sum_j \Delta J_j^i \,\mathbf{1}_{T_j^i \le t}, \tag{2}$$

the log price X^i is a diffusion and N is a vector of order flow or announcement counting processes. Asking whether X^i is locally independent of N^j that is, whether order type j genuinely drives the price is exactly the test I need. No valid test for this currently exists.

2.3 Related work

The literature on Granger causality in continuous time is sparse. Most applied work on high-frequency causality uses reduced-form VARs estimated on discretised data (e.g., Hasbrouck and Seppi, 2001), which cannot test conditional local independence. The Hawkes process literature models cross-excitation between point processes (e.g., Embrechts and Hawkes, 2011) but does not extend to mixed diffusion counting systems. On the theory side, Mogensen (2023) develops instrumental processes for LIGs an idea I intend to use for identification in some of the empirical applications. Pre averaging tools for handling microstructure noise in high frequency data are developed in Jacod et al. (2009) and Hounyo et al. (2023), which I plan to draw on heavily in the theoretical work.

3 Proposed Research Programme

My project has two theoretical aims and one empirical aim, which I expect to develop into three papers over the course of the PhD.

3.1 Paper 1: Extending the X-LCT to diffusion targets

The natural extension and a fundamental obstacle

The most natural diffusion analogue of the LCM replaces the stochastic integral against $\mathrm{d}M_s$ in equation (1) with optional quadratic covariation against the counting process:

$$\gamma_t^X = \mathbb{E}\left[\int_0^t G_s^X \,\mathrm{d}[G^X,\, N^\alpha]_s\right], \tag{3}$$

where $G_t^X = X_t - X_0 - \int_0^t \widehat{\Pi}[\mu_s \mid \mathcal{F}_s^{-\alpha}]\,\mathrm{d}s$ is the drift residual after projecting out the influence of N^α. Under the null of conditional local independence, one would want $\gamma_t^X = 0$.

There is, however, a non trivial problem with this approach that I have not yet resolved. If X is a purely continuous Itô process, then $[X, N]_t = 0$ identically, since a continuous finite variation process and a pure jump process have zero quadratic covariation with each other. This means that γ_t^X as defined in (3) would be zero regardless of whether N^α actually drives X the measure degenerates under both the null and the alternative, and no useful CLT can be built on it.

For the approach to work, there needs to be a non trivial interaction between X and N that the estimator can actually detect. One route that I think is promising is to work with pre averaged increments of X rather than the raw increments. Pre averaging with a window of size $B = \lfloor C\sqrt{n} \rfloor$ (Jacod et al., 2009) smooths the process in a way that smears the counting-process increments into a locally continuous object, potentially creating a detectable cross-term even when X is continuous. Whether this yields a non degenerate asymptotic variance under the null and whether the resulting statistic still converges to

a Brownian bridge limit is, as far as I know, an open question. Establishing this would be the main theoretical contribution of Paper 1.

How I think the proof would go, if the degeneracy can be handled

Assuming the pre averaging route resolves the degeneracy, the proof strategy would adapt the six-term decomposition of Christgau et al. (2023), Theorem 4.6, with three new components: replacing the counting martingale increment $\mathrm{d}M_s$ with $\sigma_s \, \mathrm{d}W_s$ and reworking the associated quadratic variation calculations; showing that pre averaging reduces the microstructure bias from $O(\delta_n^{1/2})$ to $O_p(n^{-3/4})$; and verifying that the dependent wild bootstrap of Hounyo et al. (2023) continues to deliver finite sample validity in this setting. The projection of the drift onto $\mathcal{F}^{-\alpha}$ would be estimated using the historical functional linear model of Malfait and Ramsay (2003), with estimation error controlled via a DML product rate condition similar to that in Christgau et al. (2023).

I am aware that this is genuinely hard, and I want to be honest that I do not yet know whether the pre averaging route will work as hoped. But I think the ingredients are in place, and the structure of the problem is similar enough to the existing pre averaging CLT literature that a solution seems within reach.

3.2 Paper 2: Extension to heavy-tailed volatility

A natural follow-up, and one I intend to pursue only once Paper 1 is on solid ground, is to ask what happens when log-prices are driven by a Lévy process with regularly varying tails at index $\alpha_L \in (1, 2)$. In this case, second moments may not exist, and the Gaussian CLT that underpins Paper 1 breaks down. The LCM would need to be replaced by an α_L covariation functional, and the limit would be an $(\alpha_L/2)$ stable process rather than a Brownian bridge. The extremal CLT of Rønn-Nielsen and Stehr (2022) looks like the right tool here, but combining it with the pre averaging construction of Paper 1 is not straightforward and would require new arguments.

Paper 2 would also address the question of LIG identifiability when the system is partially observed that is, when some processes are latent. This is important for the empirical applications, where we can never observe all relevant market participants. Mogensen (2025) characterises the identifiable equivalence class for pure counting process systems; I would aim to extend this to the mixed diffusion-counting case.

I expect Paper 2 to be the harder of the two theoretical aims, and I hold it at arms length for now.

3.3 Paper 3: Empirical applications

If the theory in Papers 1 and 2 can be developed, I want to apply it to a set of empirical questions that are currently difficult to answer well. I describe five candidate applications

below. For each, I try to be clear about what the method would add and what I would still need to do to make the application convincing.

Application I: Volatility spillovers across trading sessions

A large literature studies co-movement in volatility across markets (e.g., Diebold and Yilmaz, 2012). Using the triplet (RV_{SP}, RV_{DAX}, RV_{NK}) of five minute realised variances from the Oxford Man Realised Library, I would apply the X-LCT to recover the directed LIG of volatility spillovers across the US, European, and Asian trading sessions. The advantage over standard Granger VAR is that local independence testing does not require covariance stationarity, which realised variance series routinely violate. I see this mainly as a validation exercise a setting where we have reasonable prior expectations about the results and can benchmark against the existing literature.

Application II: Central bank announcements and FX dynamics

A natural application of the mixed system X-LCT is to ask whether central bank announcement events genuinely precede exchange rate movements, or whether the apparent relationship is driven by a latent facto say, a common response to macroeconomic news that moves both policy and prices. I plan to test this for ECB, Fed, and BoJ announcement counting processes and the corresponding EUR/USD, GBP/USD, and USD/JPY diffusions. If the LIG recovers a directed causal structure, I would map it to impulse response functions using the local spectrum framework of Andersen and Varneskov (2021), which is designed for persistent systems.

Application III: Rare earth export controls and commodity currencies

China controls approximately 70% of global rare earth mining and over 90% of refining capacity (USGS, 2024). Between 2023 and 2025, MOFCOM imposed at least 27 documented export restriction events (Table 1). These events form a natural counting process, and the question is whether they carry directed causal edges to commodity-linked exchange rates AUD/USD, CAD/USD, CLP/USD, ZAR/USD or whether the co movement is spurious, driven by a latent geopolitical factor that affects both Chinese policy and currency markets simultaneously.

Table 1: Selected MOFCOM rare earth export restriction events, 2023-2025 (27 total; source: Global Trade Alert).

Date (effective)	Material(s)	Mechanism
1 Aug 2023	Gallium, Germanium	Export licence required
1 Dec 2023	Graphite	End-user licensing
15 Sep 2024	Antimony, superhard materials	Licence + end-use review
3 Dec 2024	Ga, Ge, Sb (to United States)	Complete ban
4 Feb 2025	Tungsten, Tellurium	Export controls
4 Apr 2025	Tb, Dy, Gd + heavy REEs	Case-by-case licensing

Identification is the main challenge here. A promising strategy is to use US China bilateral diplomatic event counts as an instrumental process in the sense of Mogensen (2023). The idea is that diplomatic events should predict the timing of export restrictions without directly affecting currency markets through any other channel a restriction that is itself testable using the X-LCT, which I find reassuring. The clustering of export control events in time suggests a bivariate Hawkes process model, which introduces additional estimation steps. Whether the Hawkes residualisation that is, replacing raw event increments with residuals from the fitted Hawkes model can be cleanly integrated into the X-LCT without distorting its size is something I would need to verify carefully before treating the results as reliable.

Application IV: Price discovery and order flow

Which order types actually drive prices in a limit order book? This is one of the oldest questions in market microstructure (Glosten and Milgrom, 1985; Hasbrouck and Seppi, 2001), and it has proven surprisingly hard to answer cleanly, in large part because the log mid-price and order-flow processes are all driven partly by the same latent informed-trader signal. Applying the X-LCT to the system (X_t^{mid}, N_t^{buy}, N_t^{sell}, N_t^{cancel}, N_t^{trade}) would allow me to test for the first time with a method that has valid size control in this setting whether each order type carries a directed causal edge to the price, or only a bidirected edge pointing to a shared latent factor. I plan to use LOBSTER limit order book data for S&P 500 constituents, and to control family-wise error rates across the multiple hypotheses using Beyhum and Striaukas (2024).

Scheduled liquidity events such as index rebalancings and options expiry could serve as instruments for trade flow, helping to separate informed from uninformed order activity. Whether this instrument is strong and valid enough in practice is an empirical question I would need to address.

Application V: Does knowing the causal structure help prediction?

A natural question is whether restricting a predictive model to its estimated causal parents rather than the full covariate set improves out-of-sample performance. This is genuinely interesting but also genuinely hard to answer, and I want to be careful not to oversell it.

The honest answer is that causal restriction need not improve prediction in general. If the training and test distributions are the same, a well regularised model on the full covariate set should do at least as well as one restricted to causal parents. The case where the causal model has an advantage is when the dependence structure shifts between training and deployment some correlations that were present in training break down in testing, and the causal model is more robust because it relies only on relationships that are structurally stable. This is plausible in financial data, where structural breaks are common, but it would need to be demonstrated empirically rather than assumed. In particular, I would

need to: estimate the LIG on a training window; restrict the predictive model to the recovered causal parent set; and compare out-of-sample performance across subperiods with varying market regimes, checking whether the performance advantage of the causal model is largest precisely in subperiods that differ most from training. I see this as an ambitious target rather than a core deliverable, and I am aware that it may not produce clean results.

4 Data

For the empirical applications, I plan to use the following data sources. Tick-level FX spot rates and equity index futures from Refinitiv Tick History (CBS subscription); five-minute realised variance measures from the Oxford-Man Realised Library (21 assets, 2000–2023, freely available at https://realized.oxford-man.ox.ac.uk); central bank announcement timestamps from the Federal Reserve, ECB, and Bank of Japan; export restriction event timestamps from Global Trade Alert and MOFCOM; diplomatic event timestamps from the US State Department and the CFR Global Conflict Tracker; full NASDAQ limit order book records from LOBSTER (academic licence, 2010–2024); millisecond trade and quote records from TAQ via WRDS (CBS subscription); earnings announcement timestamps from IBES via WRDS; and rare earth spot prices from Refinitiv Datastream.

Pre-processing will follow standard practice in the high frequency literature: five minute previous tick sampling; pre averaging with window $B = \lfloor \sqrt{n} \rfloor$ (Jacod et al., 2009); bipower variation jump separation (Barndorff-Nielsen and Shephard, 2004); and UTC alignment of all event timestamps.

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